Visual Complexity Scoring: Prompt Version Comparison (V0–V3)

Model: Claude Opus 4.6 | **Evaluation Set:** 43 non-anchor ground-truth images | **Date:** April 2026

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1. Experiment Overview

We evaluated four progressively enriched prompt designs for predicting visual complexity (VC) scores of data visualization images. Each version adds a layer of guidance to the LLM, testing how definition clarity, dimensional decomposition, anchor-based calibration, and weighted scoring affect prediction quality.

Version	Short Name	Key Additions
V0	Zero-shot baseline	VC definition only; returns `vc_score` + explanation
V1	Dimension-enriched	Adds 7 dimension definitions; returns 7 sub-scores + `vc_score` (arithmetic mean)
V2	Anchor-calibrated	VC definition + 3 few-shot anchor images (vc_score only); no dimensions
V3	Full pipeline	Dimensions + calibration guidance + weighting instructions + 3 anchors (full scores)

Ground truth comes from 46 human-rated images. Three anchor images (VisC.503.6.png at 0.22, InfoVisJ.619.17.png at 0.54, InfoVisJ.1149.6(1).png at 0.95) are excluded from evaluation, leaving **43 non-anchor images** for fair comparison.

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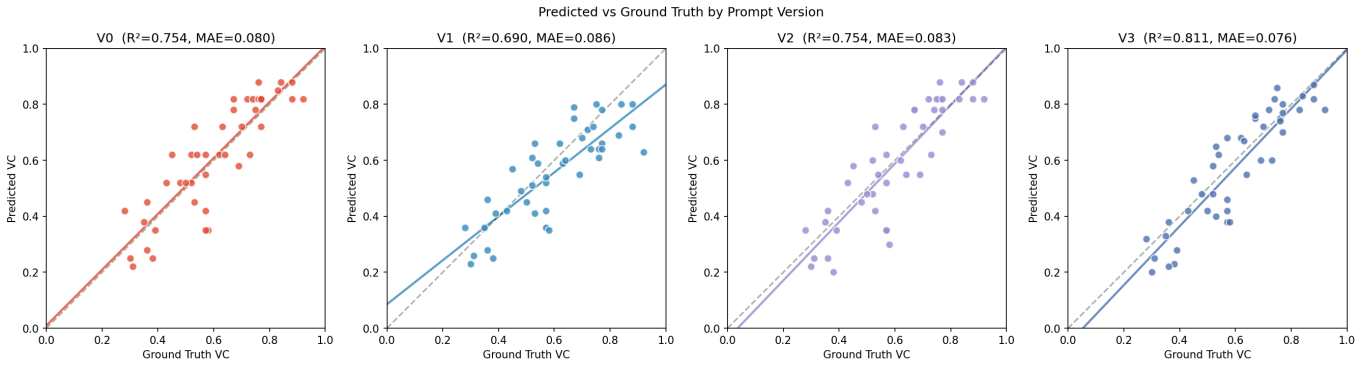
2. Summary Results

2.1 Performance Metrics

Metric	V0	V1	V2	V3
Pearson *r*	0.868	0.830	0.868	**0.901**
Spearman *ρ*	0.866	0.816	0.865	**0.896**
R^2	0.754	0.690	0.754	**0.811**
MAE	0.080	0.086	0.083	**0.076**
RMSE	0.098	0.107	0.103	**0.091**
Bias	**+0.009**	−0.043	−0.012	−0.025
p (Pearson)	< 0.0001	< 0.0001	< 0.0001	< 0.0001

2.2 Distributional Statistics

Statistic	GT	V0	V1	V2	V3
Mean	0.601	0.610	0.557	0.589	0.575
Std	0.174	0.200	0.164	0.209	0.202
Min	0.28	0.22	0.23	0.20	0.20
Max	0.92	0.88	0.80	0.88	0.87



3. Analysis and Observations

3.1 V3 is the clear winner — but the margin tells a nuanced story

V3 achieves the best performance on every regression metric: the highest Pearson correlation (0.901), highest R^2 (0.811), and lowest MAE (0.076) and RMSE (0.091). It explains **81% of the variance** in human judgments. This confirms that the full prompt — combining dimensional decomposition, calibration guidance, weighting instructions, and richly annotated few-shot anchors — produces the most human-aligned predictions.

However, the improvement from V0/V2 ($R^2 \approx 0.754$) to V3 ($R^2 = 0.811$) is meaningful but not dramatic. The model already has strong intuition about visual complexity with minimal prompting.

3.2 The surprising competence of V0 (zero-shot)

V0 — with nothing but a one-paragraph VC definition — achieves $r = 0.868$ and $R^2 = 0.754$, essentially matching V2 (which uses three anchor images). This is a striking result. It suggests that Claude Opus 4.6 has a strong pre-existing concept of visual complexity that aligns well with human ratings, even without any guidance on what dimensions matter or what the scale endpoints look like.

V0 also has the **lowest bias** (+0.009), meaning its predictions are almost perfectly centered on the GT distribution. This is notable because all other versions exhibit negative bias (systematic under-prediction).

3.3 V1 is the weakest — dimensions without calibration can hurt

V1 adds 7 dimension definitions but no calibration guidance or anchors. Rather than improving over V0, it actually *degrades* performance (r drops from 0.868 to 0.830; R^2 from 0.754 to 0.690). V1 also shows the strongest negative bias (−0.043) and a compressed range (max = 0.80 vs. GT max = 0.92).

Why dimensions alone hurt. When the model is asked to score 7 sub-dimensions and average them, two problems emerge:

1. **Range compression**: The arithmetic mean of 7 scores naturally drifts toward the center. A visualization with very high `data_density` (0.85) but low `text_annotation` (0.05) gets pulled toward an average that under-represents its true perceptual complexity.
2. **Equal weighting is wrong**: Not all dimensions contribute equally to perceived complexity. In V1, `text_annotation` — which is often near-zero for code-heavy scientific visualizations — drags the overall score down. V3 solves this with explicit weighting instructions.

3.4 V2 matches V0 — anchors without dimensions add little

V2 provides three anchor images with vc_score only (no dimension breakdown). Its metrics ($r = 0.868$, $R^2 = 0.754$) are virtually identical to V0's. The anchors do modestly improve calibration (bias: −0.012 vs. +0.009), but the effect is small.

This suggests that for a model with strong zero-shot VC intuition, simply showing a few scored examples adds marginal value. The anchors become more useful when paired with the dimensional framework (as in V3), because they calibrate not just the overall score but the *interpretation of each dimension*.

3.5 The bias-accuracy trade-off

Version	Bias	MAE
V0	+0.009 (best)	0.080
V2	−0.012	0.083
V3	−0.025	**0.076** (best)
V1	−0.043 (worst)	0.086 (worst)

V0 has the most unbiased predictions on average, but V3 has the lowest error per image. V3's small negative bias (−0.025) indicates it slightly under-predicts overall, but its image-level accuracy is superior. This is because V3 better *ranks* images (highest Spearman ρ) and has tighter per-image residuals, even though it is slightly conservative on average.

The calibration guidance ("push scores up") in V3 partially counteracts the downward pull of the text_annotation dimension but doesn't fully eliminate it — hence the residual negative bias.

3.6 Where V3 gains most: mid-range and high-complexity images

Looking at the per-image data, V3's advantage is most pronounced for images in the 0.55–0.80 GT range — moderately complex visualizations where the model needs nuanced judgment. V0 tends to over-predict in this range (positive bias), while V1 under-predicts. V3 strikes the best balance.

For very low complexity images (GT < 0.35), all versions perform similarly — the model consistently recognizes simple charts. For very high complexity images (GT > 0.85), V3 and V2 both perform well, likely because the high-complexity anchor (0.95) provides a strong calibration reference.

3.7 Standard deviation: V0 and V3 spread more than GT

V0 (std = 0.200) and V3 (std = 0.202) both exhibit wider spread than the GT (std = 0.174), while V1 (std = 0.164) is slightly compressed. V2 (std = 0.209) has the widest spread. The versions with anchors (V2, V3) tend

to use more of the [0, 1] range, which is generally desirable — but V2 may slightly overshoot.

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4. Key Takeaways

1. **Claude Opus 4.6 has strong zero-shot VC intuition.** Even with minimal prompting (V0), it achieves r = 0.87 against human ratings. This suggests the concept of "visual complexity" is well-represented in the model's training.

2. **Dimensions without calibration hurt.** V1 shows that asking the model to decompose complexity into sub-dimensions and average them — without guidance on weighting or scale — degrades performance. The arithmetic mean creates range compression and over-weights low-signal dimensions.

3. **Anchors alone add little.** V2's three anchor examples provide minimal lift over V0. The model doesn't need to be shown examples to understand the scale — it needs to be told *how to weight* different aspects of complexity.

4. **The full V3 prompt works best.** Combining dimensions, weighting, calibration guidance, and richly annotated anchors yields R^2 = 0.81 — a 7.5% relative improvement over the zero-shot baseline. Each component contributes, but none alone is sufficient.

5. **Residual negative bias persists.** All non-baseline versions (V1–V3) under-predict VC, with V3's bias at −0.025. This may be inherent to multi-dimensional scoring (the averaging effect) and could potentially be addressed with post-hoc bias correction or further tuning of the weighting instructions.

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5. Implications for Pipeline Design

- For **production use**, V3 is recommended — it provides the highest accuracy, dimensional interpretability, and acceptable bias.
- For **rapid prototyping** or when anchor images are unavailable, V0 is a surprisingly strong fallback.
- The **weighting instruction** in V3 is a critical component. Future work could experiment with learned weights or letting the model determine weights per image.
- **Anchor selection** matters less for overall accuracy than **prompt structure** — the jump from V2 to V3 comes from calibration guidance and weighting, not from adding more anchors.

Appendix A: V0 Prompt (Zero-Shot Baseline)

System Prompt

```
You are a visual complexity (VC) scoring expert for data visualizations.

**Visual Complexity** measures the amount of detail, intricacy, and
perceptual/cognitive demand
of a visualization image. It is scored on a 0–1 scale where 0 = minimal complexity
and
1 = maximum complexity.

You will receive a single visualization image with NO accompanying text or
```

```
comments.
Score it purely from what you see.

## Output Format
Return ONLY valid JSON (no markdown fences, no explanation outside JSON):
{
  "vc_score": <float 0-1>,
  "explanation": "<2-3 sentence justification>"
}
```

User Prompt

```
Score the visual complexity of this visualization image.
```

Message Structure

```
system:  [SYSTEM_PROMPT]
messages:
  user:  [image] + "Score the visual complexity of this visualization image."
```

No anchors. No dimensions. Single-turn, single-image.

Appendix B: V1 Prompt (Dimension-Enriched Zero-Shot)

System Prompt

```
You are a visual complexity (VC) scoring expert for data visualizations.

**Visual Complexity** measures the amount of detail, intricacy, and
perceptual/cognitive demand
of a visualization image. It is scored on a 0–1 scale where 0 = minimal complexity
and
1 = maximum complexity.

You will receive a single visualization image with NO accompanying text or
comments.
Score it purely from what you see.

## Scoring Dimensions (each 0–1)

1. **data_density** — The perceived amount, richness, or depth of data content.
Considers
   information volume, element quantity, and visual clutter/overlap.
2. **visual_encoding** — The variety, type, and complexity of graphical forms
(shapes, lines,
   marks) and how spatial layout, scale, and encoding interpretability contribute
```

```
to complexity.
3. **text_annotation** — The quantity and density of text elements (titles, axis
labels, legends,
   captions, annotations, in-chart labels).
4. **domain_schema** — Whether specialized domain knowledge is needed, including
dimensionality
   (2D/3D), structural complexity, and abstraction level.
5. **color_symbol** — Range, variety, and arrangement of colors, plus use of
symbols, textures,
   and non-color graphical markers.
6. **aesthetic_order** — How visually cluttered, dense, or disordered the layout
appears.
   Higher = more cluttered/overwhelming. A clean minimal layout scores low; a
crowded layout
   with overlapping elements scores high.
7. **cognitive_load** — Overall ease or difficulty of interpreting the
visualization. Considers
   interpretive difficulty, semantic clarity, and processing time/effort.

## Overall vc_score
The overall vc_score is the average of the 7 dimension scores, reflecting the
image's holistic
visual complexity.

## Output Format
Return ONLY valid JSON (no markdown fences, no explanation outside JSON):
{
  "data_density": <float 0-1>,
  "visual_encoding": <float 0-1>,
  "text_annotation": <float 0-1>,
  "domain_schema": <float 0-1>,
  "color_symbol": <float 0-1>,
  "aesthetic_order": <float 0-1>,
  "cognitive_load": <float 0-1>,
  "vc_score": <float 0-1>,
  "data_density_explanation": "<1 sentence>",
  "visual_encoding_explanation": "<1 sentence>",
  "text_annotation_explanation": "<1 sentence>",
  "domain_schema_explanation": "<1 sentence>",
  "color_symbol_explanation": "<1 sentence>",
  "aesthetic_order_explanation": "<1 sentence>",
  "cognitive_load_explanation": "<1 sentence>",
  "summary": "<2-3 sentence overall description>"
}
```

User Prompt

```
Score the visual complexity of this visualization image.
```

Message Structure

```
system:  [SYSTEM_PROMPT]
messages:
  user:  [image] + "Score the visual complexity of this visualization image."
```

No anchors. Dimensions with equal weighting (arithmetic mean). Single-turn, single-image.

Appendix C: V2 Prompt (Anchor-Calibrated, No Dimensions)

System Prompt

Same as V0:

```
You are a visual complexity (VC) scoring expert for data visualizations.

**Visual Complexity** measures the amount of detail, intricacy, and
perceptual/cognitive demand
of a visualization image. It is scored on a 0–1 scale where 0 = minimal complexity
and
1 = maximum complexity.

You will receive a single visualization image with NO accompanying text or
comments.
Score it purely from what you see.

## Output Format
Return ONLY valid JSON (no markdown fences, no explanation outside JSON):
{
  "vc_score": <float 0-1>,
  "explanation": "<2-3 sentence justification>"
}
```

Anchor Messages (3 few-shot examples)

Each anchor is presented as a user→assistant turn pair before the target image:

Anchor 1 — Low complexity (GT = 0.22)

- Image: `VisC.503.6.png`
- User: `[image]` + `"Score the visual complexity of this visualization image."`
- Assistant: `{"vc_score": 0.22, "explanation": "Anchor image scored at 0.22."}`

Anchor 2 — Mid complexity (GT = 0.54)

- Image: `InfoVisJ.619.17.png`
- User: `[image]` + `"Score the visual complexity of this visualization image."`
- Assistant: `{"vc_score": 0.54, "explanation": "Anchor image scored at 0.54."}`

Anchor 3 — High complexity (GT = 0.95)

- Image: `InfoVisJ.1149.6(1).png`
- User: `[image]` + `"Score the visual complexity of this visualization image."`
- Assistant: `{"vc_score": 0.95, "explanation": "Anchor image scored at 0.95."}`

User Prompt (target image)

```
Score the visual complexity of this visualization image.
```

Message Structure

```
system:  [SYSTEM_PROMPT, cache_control: ephemeral]
messages:
  user:      [anchor_1_image] + "Score the visual complexity..."
  assistant: {"vc_score": 0.22, "explanation": "Anchor image scored at 0.22."}
  user:      [anchor_2_image] + "Score the visual complexity..."
  assistant: {"vc_score": 0.54, "explanation": "Anchor image scored at 0.54."}
  user:      [anchor_3_image] + "Score the visual complexity..."
  assistant: {"vc_score": 0.95, "explanation": "Anchor image scored at 0.95."}
[cached]
  user:      [target_image] + "Score the visual complexity..."
```

Appendix D: V3 Prompt (Full Pipeline)

System Prompt

```
You are a visual complexity (VC) scoring expert for data visualizations.

**Visual Complexity** measures the amount of detail, intricacy, and
perceptual/cognitive demand
of a visualization image. It is scored on a 0–1 scale where 0 = minimal complexity
and
1 = maximum complexity.

You will receive a single visualization image with NO accompanying text or
comments.
Score it purely from what you see.

## Calibration Guidance
Use the full 0–1 range. Do NOT cluster scores conservatively toward the center.
- A plain single bar chart or pie chart with < 5 elements → vc_score ≈ 0.25–0.40
- A standard scatterplot, grouped bar chart, or line chart → vc_score ≈ 0.45–0.60
- A multi-panel dashboard, dense network, or multi-encoding composite → vc_score ≈
0.70–0.85
- An extremely dense multi-view system with many coordinated panels → vc_score ≈
0.85–0.95
IMPORTANT: If you are unsure between two scores, always choose the higher one.
```

```
Most scorers underestimate — push scores up.

## Scoring Dimensions (each 0–1)

1. **data_density** — The perceived amount, richness, or depth of data content.
Considers
   information volume, element quantity, and visual clutter/overlap.
2. **visual_encoding** — The variety, type, and complexity of graphical forms
(shapes, lines,
   marks) and how spatial layout, scale, and encoding interpretability contribute
to complexity.
3. **text_annotation** — The sheer quantity and density of text elements (titles,
axis labels,
   legends, captions, annotations, in-chart labels). Score based on volume only:
more text =
   higher score, regardless of legibility.
4. **domain_schema** — Whether specialized domain knowledge is needed, including
dimensionality
   (2D/3D), structural complexity, and abstraction level.
5. **color_symbol** — Range, variety, and arrangement of colors, plus use of
symbols, textures,
   and non-color graphical markers.
6. **aesthetic_order** — How visually cluttered, dense, or disordered the layout
appears.
   Higher = more cluttered/overwhelming. A clean minimal layout scores low; a
crowded layout
   with overlapping elements scores high.
7. **cognitive_load** — Overall ease or difficulty of interpreting the
visualization. Considers
   interpretive difficulty, semantic clarity, and processing time/effort.

## Overall vc_score
The overall vc_score is a weighted average reflecting the image's holistic visual
complexity.
Weight dimensions as follows:
- **High weight**: data_density, visual_encoding, domain_schema, cognitive_load
  (these drive VC the most)
- **Medium weight**: color_symbol, aesthetic_order
- **Low weight**: text_annotation (text volume alone is a weak VC signal)
The vc_score should be consistent with (but not necessarily the arithmetic mean
of) the
7 dimension scores. A low text_annotation score should NOT substantially pull down
the
overall vc_score.

## Output Format
Return ONLY valid JSON (no markdown fences, no explanation outside JSON):
{
  "data_density": <float 0-1>,
  "visual_encoding": <float 0-1>,
  "text_annotation": <float 0-1>,
  "domain_schema": <float 0-1>,
  "color_symbol": <float 0-1>,
  "aesthetic_order": <float 0-1>,
```

```
  "cognitive_load": <float 0-1>,
  "vc_score": <float 0-1>,
  "data_density_explanation": "<1 sentence>",
  "visual_encoding_explanation": "<1 sentence>",
  "text_annotation_explanation": "<1 sentence>",
  "domain_schema_explanation": "<1 sentence>",
  "color_symbol_explanation": "<1 sentence>",
  "aesthetic_order_explanation": "<1 sentence>",
  "cognitive_load_explanation": "<1 sentence>",
  "summary": "<2-3 sentence overall description>"
}
```

Anchor Messages (3 few-shot examples with full dimension scores)

Each anchor is presented as a user→assistant turn pair with complete dimension breakdowns:

Anchor 1 — Low complexity: `VisC.503.6.png` **(GT = 0.22)**

```
{
  "data_density": 0.15,
  "visual_encoding": 0.10,
  "text_annotation": 0.20,
  "domain_schema": 0.15,
  "color_symbol": 0.10,
  "aesthetic_order": 0.10,
  "cognitive_load": 0.20,
  "vc_score": 0.22,
  "data_density_explanation": "Only 3 bars with error bars; very sparse data
content.",
  "visual_encoding_explanation": "Standard vertical bars with simple hatching;
minimal encoding variety.",
  "text_annotation_explanation": "Basic axis labels and a title present but
minimal text overall.",
  "domain_schema_explanation": "Generic bar chart format; no specialized domain
knowledge needed.",
  "color_symbol_explanation": "Monochrome black hatched bars on white; no color
variety at all.",
  "aesthetic_order_explanation": "Clean, well-organized layout with ample
whitespace.",
  "cognitive_load_explanation": "Immediately interpretable; very low effort to
understand.",
  "summary": "A simple 3-bar chart with black hatching and error bars. Extremely
low visual complexity across all dimensions."
}
```

Anchor 2 — Mid complexity: `InfoVisJ.619.17.png` **(GT = 0.54)**

```
{
  "data_density": 0.55,
```

```
  "visual_encoding": 0.50,
  "text_annotation": 0.45,
  "domain_schema": 0.60,
  "color_symbol": 0.50,
  "aesthetic_order": 0.45,
  "cognitive_load": 0.55,
  "vc_score": 0.54,
  "data_density_explanation": "Moderate number of data points in a 2D scatterplot
with some grouping structure visible.",
  "visual_encoding_explanation": "Points plotted in a biplot layout with
directional arrows; moderate encoding complexity.",
  "text_annotation_explanation": "Axis labels and some point labels present;
moderate text volume.",
  "domain_schema_explanation": "Biplot/PCA projection requires some statistical
familiarity.",
  "color_symbol_explanation": "A few colors distinguish groups; moderate palette
diversity.",
  "aesthetic_order_explanation": "Reasonably organized but arrows and overlapping
labels add mild clutter.",
  "cognitive_load_explanation": "Requires understanding of biplots and PCA;
moderate interpretation effort.",
  "summary": "A PCA biplot with directional arrows and grouped points. Moderate
complexity requiring some statistical knowledge."
}
```

Anchor 3 — High complexity: `InfoVisJ.1149.6(1).png` **(GT = 0.95)**

```
{
  "data_density": 0.90,
  "visual_encoding": 0.85,
  "text_annotation": 0.80,
  "domain_schema": 0.95,
  "color_symbol": 0.70,
  "aesthetic_order": 0.85,
  "cognitive_load": 0.95,
  "vc_score": 0.95,
  "data_density_explanation": "Multiple coordinated panels (7 views) each showing
dense text and data; extremely high information volume.",
  "visual_encoding_explanation": "Mixed encoding types across panels: lists,
highlighted text, bar-like indicators, tag clouds; high variety.",
  "text_annotation_explanation": "Extensive text content across all panels; the
visualization IS largely text-based with highlights and annotations.",
  "domain_schema_explanation": "Specialized text analysis/collation tool; requires
understanding of multi-view coordinated systems.",
  "color_symbol_explanation": "Multiple highlight colors for text segments;
moderate color use with functional meaning.",
  "aesthetic_order_explanation": "Complex multi-panel layout with varied panel
sizes; visually dense and somewhat overwhelming.",
  "cognitive_load_explanation": "Must track relationships across 7 coordinated
views with dense text; very high cognitive demand.",
  "summary": "A multi-panel text collation tool with 7 coordinated views showing
highlighted text passages, bar indicators, and tag clouds. Extremely high visual
```

```
  complexity requiring sustained multi-view comparison."
  }
```

User Prompt (target image)

```
Score the visual complexity of this visualization image.
```

Message Structure

```
system:  [SYSTEM_PROMPT, cache_control: ephemeral]
messages:
  user:      [anchor_1_image] + "Score the visual complexity..."
  assistant: {full 15-field JSON for anchor 1}
  user:      [anchor_2_image] + "Score the visual complexity..."
  assistant: {full 15-field JSON for anchor 2}
  user:      [anchor_3_image] + "Score the visual complexity..."
  assistant: {full 15-field JSON for anchor 3}  [cached]
  user:      [target_image] + "Score the visual complexity..."
```

Appendix E: Experimental Design Notes

- **Model**: Claude Opus 4.6 (Anthropic), accessed via `anthropic` Python SDK
- **All versions use the same user-facing prompt**: `"Score the visual complexity of this visualization image."`
- **Temperature**: Default (not explicitly set)
- **max_tokens**: V0/V2 = 800; V1/V3 = 1500
- **Prompt caching**: V2 and V3 use `cache_control: {"type": "ephemeral"}` on the system prompt and last anchor assistant message to reduce latency/cost for batch scoring
- **Anchor images** are drawn from the 46-image GT set. They are excluded from the 43-image evaluation to prevent data leakage
- **Ground truth**: NormalizedVC scores from human annotations, range [0.22, 0.95], mean = 0.601, std = 0.174